perplexity

Healthcare Patient Management System (HPMS) with Facial Recognition: Product Requirements Document

Executive Summary

The Healthcare Patient Management System (HPMS) is a full-stack application designed to revolutionize patient tracking and medical record management through facial recognition technology. Leveraging Django (backend), React/Next.js (frontend), Neon (serverless PostgreSQL), and Shadcn UI with V0's AI capabilities, this system ensures seamless patient identification, cross-provider data sharing, and HIPAA-compliant security. The integration of biometric authentication with EHR management addresses critical gaps in care continuity while modernizing healthcare workflows[1] [2] [3] [4].

1. Objectives

1.1 Primary Goals

- Implement real-time facial recognition for instant patient identification across healthcare facilities
- Create centralized EHR accessible to authorized providers via secure API endpoints
- Enable automatic retrieval of medical history during patient visits through biometric matching
- Ensure GDPR/HIPAA compliance for biometric and health data storage[3] [5]

1.2 Key Performance Indicators

- 99.9% facial recognition accuracy under diverse lighting/skin tone conditions
- <500ms response time for patient record retrieval via API
- Zero data breaches through E2E encryption and role-based access

2. Scope & Features

2.1 Core Functional Modules

2.1.1 Patient Registration & Facial Enrollment

- **Biometric Capture**: Multi-angle facial image collection using React Media Pipeline
- **Liveness Detection**: AI-powered verification via Shadcn/V0 to prevent spoofing [4]
- **Demographic Intake**: Dynamic forms built with Next.js App Router
- **Consent Management**: Digital signature capture for GDPR-compliant data usage

2.1.2 Real-Time Facial Recognition Engine

- **Matching Algorithm**: 128D facial embeddings using FaceNet ML model
- **Cross-Device Optimization**: Camera calibration for mobile/desktop/webcam inputs
- **Fallback Mechanism**: QR code + OTP authentication for failed matches

2.1.3 Electronic Health Record (EHR) System

- **Unified Patient Profile**: Neon DB schema storing:

```
CREATE TABLE patients (
  facial_hash BYTEA PRIMARY KEY,
  medical_history JSONB,
  prescriptions JSONB,
  allergies TEXT[]
) USING neon_engine;
```

- **Timeline Visualization**: React Visx charts displaying treatment history
- **Clinical Decision Support**: Drug interaction alerts via Django background workers

2.2 User Portals

2.2.1 Doctor Interface (Next.js)

- Real-time patient lookup via camera/webcam
- Treatment plan builder with Shadcn form components
- Secure messaging system using WebSockets

2.2.2 Admin Dashboard (React Admin)

- Biometric data audit trails
- API usage analytics via Neon Query Metrics [3]
- RBAC configuration with Django Guardian

3. Technical Architecture

3.1 Backend Stack (Django)

- **REST API**: Django REST Framework with:

  ```
  class PatientViewSet(ModelViewSet):
    queryset = Patient.objects.using('neon').all()
    serializer_class = PatientSerializer
    permission_classes = [IsDoctor]
  ```

- **Task Queue**: Celery for async facial matching jobs
- **Security**: JWT authentication with refresh token rotation

3.2 Frontend Stack (Next.js 14)

- **UI Components**: Shadcn CLI-generated accessible elements [6] [4]

  ```
  npx shadcn@latest add face-scanner -y
  ```

- **AI Integration**: V0.dev for automated UI generation:

  > "Generate a patient dashboard with vital stats chart" → AI outputs React code

- **State Management**: Zustand for global facial match status

3.3 Database (Neon Serverless Postgres)

- **Storage Architecture**:
  - **Facial Hashes**: Neon columnar storage for fast similarity searches
  - **Medical Records**: Row-based storage with JSONB for flexibility [7] [3]
- **Disaster Recovery**: Cross-region replication via Neon's branching system

4. Non-Functional Requirements

4.1 Performance

Metric	Target
Facial Match Latency	≤300ms (p95)
API Response Time	≤500ms (p99)
Concurrent Users	10,000+

4.2 Security

- **Data Encryption**: AES-256 for biometric templates + Argon2 for hashing
- **Compliance**: HIPAA audit trails + GDPR right-to-be-forgotten implementation
- **Penetration Testing**: Monthly OWASP ZAP scans + HackerOne bug bounty

4.3 Accessibility

- WCAG 2.1 AA compliance via Shadcn's built-in a11y features[6]
- Screen reader support for all medical data visualizations

5. Development Roadmap

Phase 1: Foundation Setup (Weeks 1-4)

- ✅ Django-Neon integration using `django-neon` adapter[3]
- ✅ Next.js App Router configuration with Shadcn theme
- ✅ Facial recognition PoC with FaceNet ONNX model

Phase 2: Core Development (Weeks 5-12)

- ▯ Patient registration workflow with liveness checks
- ▯ Real-time matching API using Django Channels
- ▯ EHR schema design + Neon performance tuning

Phase 3: AI Integration (Weeks 13-16)

- ▯ V0.dev training for medical UI generation[4]
- ▯ Predictive analytics using Django ML libraries

6. Risks & Mitigation

Risk	Mitigation Strategy
Biometric bias in recognition	Implement multimodal authentication fallback
HIPAA compliance gaps	Engage healthcare legal consultants early
Neon cold-start latency	Warm connection pooling + standby replicas

7. Conclusion

HPMS represents a paradigm shift in healthcare IT by combining cutting-edge biometrics with modern web technologies. By leveraging Django's robustness, Neon's serverless scalability, and Shadcn/V0's AI-driven UI capabilities, this system addresses critical challenges in patient data management while maintaining strict compliance standards. The architecture enables future expansion into telemedicine integrations and predictive health analytics through its modular design[2] [3] [4].

Appendices
A. Detailed Neon DB Schema
B. Facial Recognition Accuracy Benchmarks
C. Shadcn UI Component Library Specifications[6] [4]

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1. https://www.interviewbit.com/blog/django-architecture/
2. https://nextbigtechnology.com/building-restful-apis-with-django-rest-framework-a-comprehensive-guide/
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4. https://dev.to/stonediggity/httpscdn-images-1mediumcommax30441o0znhwukiy1wcgmys76skqpng-39nc
5. https://lumiformapp.com/templates/software-product-requirements-document-template
6. https://ui.shadcn.com/docs/installation/next
7. https://github.com/neondatabase/neon